

Liver cancer ablation

Tumour ablation means destroying the cancer cells by applying heat, cold, electrical pulses or alcohol directly to the tumour. It can be used to treat small primary liver tumours or sometimes a secondary cancer in the liver.

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What is ablation for liver cancer?

Tumour ablation means destroying the tumour by applying heat, cold, electrical pulses or alcohol directly to it.

Primary liver cancer

For [primary liver cancer](#) (also known as hepatocellular carcinoma or HCC) ablation can be used to treat small tumours in the liver.

If you have one tumour that is only 2cm or smaller, tumour ablation can be as effective as [surgery](#).

It may be used in the following ways:

- Instead of surgery, usually for tumours smaller than 3cm. If the tumour is nearly 3cm, you might have a treatment called [trans-arterial chemoembolisation \(TACE\)](#), followed by tumour ablation a few weeks later.
- If you cannot have surgery, or if the tumour is small but very close to the outside of [the liver](#), which makes it hard to remove with surgery.
- With or after other treatments, such as surgery and TACE.

Secondary cancer in the liver

Ablation is most often used to treat [cancer that has spread to the liver](#) from the bowel or breast. But sometimes it is used for other types of cancer that have spread to the liver. It is usually used to treat small tumours when surgery is not possible. It is also sometimes done during [surgery](#). Ablation might be used in combination with other treatments, such as [chemotherapy](#). Some people have ablation more than once.

Tumour ablation can be done:

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during a [laparoscopy](#) – the ablation needle is inserted into the liver tumour through the keyhole instruments

- during open surgery, usually a [liver resection](#) – the ablation needle is put directly into the liver tumour
- as an injection with a needle directly into the liver through the skin (percutaneously).

Your liver specialist will explain whether this type of treatment may be suitable for you. If your doctor thinks tumour ablation may help you, they can refer you to a hospital that does this treatment. Tumour ablation treatments are only available in some specialist hospitals, so you may have to travel for treatment.

Getting support

We understand that having treatment can be a difficult time for people. We are here to support you. If you want to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:08088080000).
- [Chat to our specialists online](#).
- Visit our [liver cancer forum](#) to connect with people who have been affected by liver cancer, share your experience, and ask your questions.

Types of tumour ablation

Types of tumour ablation include:

- microwave ablation (MWA)
- radiofrequency ablation (RFA)
- cryotherapy
- irreversible electroporation (IRE)
- percutaneous ethanol injection (PEI).

The most common types are MWA and RFA. They use microwaves or radio waves to produce heat and destroy the tumour.

Cryotherapy produces very low temperatures, which freeze the tumour. This is less commonly done.

Irreversible electroporation uses a high voltage current to destroy tumours. It does not heat the nearby tissue, so it is useful for tumours that are close to vessels or bile ducts.

Your doctor will talk to you about the best type of tumour ablation for you.

Percutaneous ethanol injection (PEI)

This type of ablation uses alcohol and works in a different way. Percutaneous means given through the skin. The doctor injects pure alcohol (ethanol) through a needle passed through the skin into the tumour. The alcohol destroys the cancer cells.

You have this done in the scanning department. The doctor gives you a local anaesthetic to numb the area. They use an [ultrasound scan](#) to help guide the needle into the tumour.

You usually need several treatments, depending on the number of tumours and their size. If the tumour grows again, you can have the treatment again.

What happens during ablation?

You usually have ablation under a general anaesthetic. The doctor puts the ablation needle through the skin over the liver and into the centre of each tumour. They use an [ultrasound](#) or [CT scan](#) to guide them to target the tumours using ablation.

You may need to stay in hospital overnight. Afterwards, you will have 1 to 3 tiny holes in your tummy area (abdomen). These usually heal quickly. You usually have a CT scan a few weeks after ablation to see how well it has worked.

Side effects of liver cancer ablation

The side effects of tumour ablation are usually mild and may last up to a week. Your doctor or nurse will tell you more about the side effects of the type of ablation treatment you are having.

You may have pain in the liver area. You can control this by taking regular painkillers. Other side effects include:

- a fever (high temperature)
- tiredness
- feeling generally unwell.

These side effects happen because the body is getting rid of the cells that have been destroyed. Try to drink plenty of fluids and get enough rest. Your doctor or nurse may ask you to contact the hospital if your temperature:

- does not go back to normal after a few days
- goes above 38°C.

These may be signs that you have an [infection](#).

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our primary liver cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help